

TAC 368 Final Project Proposal

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Project: Omit - Text Image Redactor

Overview

Omit is a mobile app that allows users to quickly and easily censor text in screenshots or photos. The user uploads an image from their photo library, and the app automatically detects all text in the image using on device text recognition. The user can then tap on any detected text to cover it with a black rectangle, effectively redacting it. Once satisfied, the user can save the redacted image back to their photo library. This is useful for sharing screenshots or photos on social media or with others while hiding personal information such as names, phone numbers, addresses, or account details.

How It Works

When the app launches, the user is presented with a simple screen containing an "Upload Image" button. Tapping it opens the device's photo library so the user can select an image. After selection, the app runs Google ML Kit's text recognition on the image entirely on device, with no internet connection required. The recognized text regions are displayed as semi transparent blue overlays on top of the image so the user can see exactly what was detected.

Tapping any highlighted region toggles it to a solid black rectangle, marking it as redacted. Tapping again removes the redaction, giving the user full control over what gets hidden. A "Save" button at the bottom of the screen renders the final image with the black rectangles permanently drawn onto the pixels and saves it to the photo library.

Features and Requirements

The app uses images as a core part of the experience, since the entire workflow revolves around loading, annotating, and exporting images. For data persistence, the app will store a history of recently redacted images using local storage (shared preferences or a lightweight local database), so the user can revisit previous edits between sessions. The interface will be clean and focused, with appropriate use of color to distinguish between detected text (blue overlay) and redacted text (black overlay). Error handling will cover cases such as the user canceling the image picker or selecting an image with no detectable text.